Codilar_Twoxl

Magento 2.4.9 implementation guide — 2XL Home storefront

This document records everything built in the **Codilar_Twoxl** module and the **Codilar/twoxl** theme: the attributes added and where their values live in the database, the products and the colour configurable, the second regional website, the custom enquiry table, and the admin and storefront code around them.

Each section lists what was implemented, every file involved, the database tables and indexes touched, and the steps to verify it.

Platform	Magento Open Source 2.4.9 (PHP 8, MySQL, nginx)
Module	`app/code/Codilar/Twoxl`
Theme	`app/design/frontend/Codilar/twoxl`
Websites	Main Website (UAE, AED) and 2XL Home India (INR)
Storefront	`http://midhun2.test/ae_en/` | `http://midhun2.test/in_en/`
Admin	`http://midhun2.test/admin_new`
Data patches	19 applied (see Appendix B)

1. How the codebase is organised

Everything is split between one module and one theme. The rule followed throughout: **PHP logic lives in the module, markup and styling live in the theme**, and the theme's templates receive module ViewModels as layout arguments.

1.1 Module layout

```
app/code/Codilar/Twoxl/
├── etc/
│   ├── module.xml              module declaration + sequence
│   ├── di.xml                  en_IN locale, enquiry grid data source
│   ├── db_schema.xml           the custom enquiry table
│   ├── db_schema_whitelist.json  generated; lets setup:upgrade alter it
│   ├── acl.xml                 admin permission resource
│   ├── adminhtml/routes.xml    admin front name  twoxl_admin
│   ├── adminhtml/menu.xml      Catalog -> Product Enquiries
│   └── frontend/{routes,events}.xml
├── Setup/Patch/Data/           19 data patches (all schema + seed data)
├── Setup/Fixtures/images/      product images shipped with the patches
├── Model/                      ProductEnquiry, GuestWishlist, sources
├── Controller/                 storefront + Adminhtml controllers
├── Ui/Component/               grid actions column
├── ViewModel/                  objects the theme templates call
└── view/adminhtml/             ui_component XML + admin layouts
```

1.2 Theme layout

```
app/design/frontend/Codilar/twoxl/
├── requirejs-config.js         registers two JS mixins
├── Magento_Catalog/
│   ├── layout/                 PDP + category layout
│   └── templates/              gallery, price, accordion, enquiry form ...
├── Magento_ConfigurableProduct/templates/product/price/final_price.phtml
├── Magento_Cms/templates/home/ home page sections
├── Magento_Theme/              header, footer, layout
└── web/
    ├── css/source/_extend.less  imports every twoxl/_*.less partial
    ├── css/source/twoxl/        one partial per area
    └── js/                      mixins and widgets
```

1.3 Why data patches, not manual admin work

Every structural change is a **data patch** under `Setup/Patch/Data/`, applied by `bin/magento setup:upgrade`. This makes the whole setup reproducible on any environment and keeps it in version control. Magento records each applied patch in the `patch_list` table and never re-runs it.

Two consequences worth knowing:

Patches are append-only	Once applied, a patch never runs again. To change something, write a *new* patch — editing an applied one has no effect on existing environments.
Moving a patch needs an alias	Renaming or relocating a patch class makes Magento treat it as new and re-run it. `getAliases()` declares the old class name so it is skipped. Used in section 3.

2. Product attributes

Eleven custom product attributes exist. Five pre-date this work; six were added for upholstered seating, plus new option values on two existing dropdowns.

2.1 What was implemented

Patch	Adds
`AddProductAttributes`	height_cm, width_cm, length_cm, material, care — the PDP specification rows
`AddSofaAttributes`	sofa_material, frame_material, filling_material, sofa_shape; tops up seating_capacity (3/5/7/8 Seater) and color (Beige, Ivory, Cream)
`AddTableMaterialAttribute`	table_material — relocated from a throwaway module (section 3)
`ConfigureColorAttribute`	makes the stock color attribute visible on the PDP
`ConfigureColorSwatches`	adds Red, converts color to visual swatches (section 6)

2.2 Where the values are stored

Magento uses EAV: the attribute *definition* and the attribute *value* live in different tables. The value table is chosen by the attribute's `backend_type`.

Attribute	ID	Input	Scope	Value table
`height_cm`	137	text	Store	`catalog_product_entity_varchar`
`width_cm`	138	text	Store	`catalog_product_entity_varchar`
`material`	139	text	Store	`catalog_product_entity_varchar`
`length_cm`	140	text	Store	`catalog_product_entity_varchar`
`care`	141	textarea	Store	`catalog_product_entity_text`
`seating_capacity`	142	select	Global	`catalog_product_entity_int`
`table_material`	143	select	Global	`catalog_product_entity_int`
`sofa_material`	144	select	Global	`catalog_product_entity_int`
`frame_material`	145	select	Global	`catalog_product_entity_int`
`filling_material`	146	select	Global	`catalog_product_entity_int`
`sofa_shape`	147	select	Global	`catalog_product_entity_int`
`color`	93	swatch	Global	`catalog_product_entity_int`

The supporting tables

`eav_attribute`	The definition: attribute_code, backend_type, frontend_input, backend_model.
`catalog_eav_attribute`	Catalog-specific flags, notably `is_global` (0 store, 1 global, 2 website) and the searchable / filterable flags.
`eav_attribute_option`	One row per dropdown option, giving it an option_id.
`eav_attribute_option_value`	The option's label per store (store_id 0 is the admin label).
`eav_entity_attribute`	Which attribute set and group the attribute appears in, and its sort order.
`catalog_product_entity_int`	The value for a select attribute — stores the **option_id**, not the label.

The key point for select attributes: the product row holds an integer option_id. A label like “Boucle” only exists in `eav_attribute_option_value`. Any code that sets one of these must resolve the label to its option_id first — writing the label string directly stores 0.

2.3 Option IDs currently in the database

Attribute	Options (option_id in brackets)
`color`	[4] Grey [40] Beige [41] Ivory [42] Cream [43] Red
`seating_capacity`	[5] 2 Seater [6] 4 Seater [7] 6 Seater [36] 3 Seater [37] 5 Seater [38] 7 Seater [39] 8 Seater
`table_material`	[8] Sheesham Wood [9] Teak Wood [10] Engineered Wood [11] Marble [12] Glass
`sofa_material`	[13] Boucle [14] Fabric [15] Velvet [16] Chenille [17] Linen [18] Leather [19] Faux Leather
`frame_material`	[20] Solid Wood [21] Engineered Wood [22] Plywood [23] Metal
`filling_material`	[24] High Resilience Foam [25] Memory Foam [26] Polyester Fibre [27] Feather Blend [28] Foam & Fibre
`sofa_shape`	[29] Corner Sofa [30] L-Shaped [31] U-Shaped [32] Curved [33] Modular [34] Chaise Sofa [35] Straight

2.4 How to test

In the admin: Stores → Attributes → Product, search for the code. Each should open with its options listed. On a product, Content tab → Product Details group shows the fields.

On the storefront: the values render in the Product Description accordion on the PDP.

In the database — read a select attribute back with its label resolved:

```
SELECT e.sku, a.attribute_code, ov.value AS label, v.value AS option_id
FROM catalog_product_entity e
JOIN catalog_product_entity_int v ON v.entity_id = e.entity_id
JOIN eav_attribute a ON a.attribute_id = v.attribute_id
JOIN eav_attribute_option_value ov
     ON ov.option_id = v.value AND ov.store_id = 0
WHERE a.attribute_code IN ('sofa_material','sofa_shape','color');
```

3. Consolidating the attribute patches

3.1 The problem

`table_material` was defined in a separate placeholder module, `Vendor_FurnitureAttributes`, while every other attribute lived in `Codilar_Twoxl`. Two modules, one concern.

3.2 Why a plain move would have broken it

The original patch was already applied and recorded in `patch_list`. Moving the class to a new namespace makes Magento see an *unapplied* patch and run it again. Its `addAttribute()` call would then append all five material options a second time, leaving the dropdown with duplicates.

3.3 The fix

Magento's supported mechanism is `getAliases()`. The relocated class declares its former name; `PatchApplier::isApplied()` checks each alias against `patch_list`, and on a hit records the new class name and skips `apply()` entirely.

```
public function getAliases(): array
{
    return ['Vendor\\FurnitureAttributes\\Setup\\Patch\\Data\\AddTableMaterialAttribute'];
}
```

Note that `PatchHistory::isApplied()` alone does **not** consult aliases — only the applier does. The patch was also made idempotent (it returns early if the attribute exists) so it stays safe even if the patch history is ever lost.

3.4 Files

Action	Path
Added	`Setup/Patch/Data/AddTableMaterialAttribute.php`
Deleted	`app/code/Vendor/  (entire placeholder module)`
Edited	`app/etc/config.php — removed Vendor_FurnitureAttributes`

3.5 How to test

```
php bin/magento setup:upgrade

# both names present, apply() ran only once
SELECT patch_name FROM patch_list WHERE patch_name LIKE '%TableMaterial%';

# must return exactly 5 - more means the options were duplicated
SELECT COUNT(*) FROM eav_attribute_option o
JOIN eav_attribute a ON a.attribute_id = o.attribute_id
WHERE a.attribute_code = 'table_material';
```

One inert leftover: the old module still has a row in `setup_module`. It is harmless and was left rather than hand-editing database rows.

4. Category banner attribute

4.1 What was implemented

A per-category hero image, `category_banner`, editable in the admin and rendered above the product list on category pages.

Attribute code	`category_banner`
Attribute ID	148
Entity	`catalog_category  (entity_type_id 3)`
Input / backend type	`image / varchar`
Backend model	`Magento\Catalog\Model\Category\Attribute\Backend\Image`
Scope	Store view (`is_global = 0`) — each region can show a different banner
Admin location	Catalog → Categories → **Content** group, sort order 45

4.2 Where the value is stored

`eav_attribute`	Definition, attribute_id 148
`catalog_eav_attribute`	is_global = 0
`eav_entity_attribute`	Places it in the Content group of attribute set 3
`catalog_category_entity_varchar`	**The value** — only the file name, e.g. `twoxl-sample-banner.png`
`pub/media/catalog/category/`	The image file itself on disk (not a database table)

`Category::getImageUrl('category_banner')` joins the media base URL to the stored file name at render time.

4.3 Why no custom upload controller was needed

Magento's core `catalog/category_image/upload` controller is generic: it reads `param_name` from the request, and the backend model keys off `getAttribute()->getName()`. Reusing it means uploads land in `pub/media/catalog/category` beside the stock category image, with no extra controller or ImageUploader virtual type.

4.4 Files

Layer	Path
Patch	`Setup/Patch/Data/AddCategoryBannerAttribute.php`
Admin form	`view/adminhtml/ui_component/category_form.xml`
ViewModel	`ViewModel/CategoryBanner.php`
Layout	`THEME/Magento_Catalog/layout/catalog_category_view.xml`
Template	`THEME/Magento_Catalog/templates/category/banner.phtml`
Styles	`THEME/web/css/source/twoxl/_category.less`

4.5 How to test

1. Admin → Catalog → Categories → **Corner Sofa** → Content → upload a Category Banner, Save.

2. Open `/ae_en/furniture/sofa-seating/corner-sofa` — the banner renders above the product grid. A category with no banner renders nothing (it degrades cleanly).

3. Confirm in the database:

```
SELECT v.entity_id, v.store_id, v.value
FROM catalog_category_entity_varchar v
JOIN eav_attribute a ON a.attribute_id = v.attribute_id
WHERE a.attribute_code = 'category_banner';
```

Scope gotcha: saving while a specific store view is selected writes that `store_id`, so the banner appears on that view only. Save on **All Store Views** to write `store_id = 0` and have both regions share it.

5. Catalogue and products

5.1 Category tree

```
Default Category (2)
■■■ Furniture (23)
    ■■■ Dining (30) → Dining Tables (31) → 8-Seater Dining (32)
    ■■■ Sofa & Seating (33) → Corner Sofa (34)
```

Both websites share root category 2, so one catalogue serves both regions.

5.2 Products

SKU	Name	Type	Categories
`130917`	Tresa 8 Seater Dining table	simple	23, 30, 31, 32
`127933`	Niecel Sectional Sofa	**configurable**	23, 33, 34
`127933-IVORY`	Niecel Sectional Sofa - Ivory	simple (hidden)	—
`127933-RED`	Niecel Sectional Sofa - Red	simple (hidden)	—

5.3 How images are attached

Patches copy fixture images from `Setup/Fixtures/images/` into `pub/media/import/codilar_twoxl/`, then call `addImageToMediaGallery()`. Magento disperses the final file by the first two characters of its name, so `127933_1.png` ends up at `/1/2/127933_1_1.png` — matching the live site's convention.

Gallery tables

`catalog_product_entity_media_gallery`	One row per image file (value_id, value = path)
`..._media_gallery_value`	Per-store position, label and disabled flag
`..._media_gallery_value_to_entity`	Links an image to a product
`catalog_product_entity_varchar`	The role attributes: image, small_image, thumbnail, swatch_image

5.4 How to test

```
/ae_en/tresa-8-seater-dining-table-130917
/ae_en/niecel-sectional-sofa-127933
/ae_en/furniture/sofa-seating/corner-sofa
```

Each PDP should show name, SKU, price with the discount badge, the gallery, and the specification accordion. Hidden child products must **not** appear in category listings.

6. Colour configurable with swatches

6.1 What was implemented

SKU `127933` was converted from a simple product into a **configurable** keyed on `color`, keeping its URL key, categories, description and specification attributes. Two hidden simple children carry their own price and their own gallery.

SKU	Colour	Price (AED)	Special	Visibility	Images
`127933`	— parent	—	—	Catalog, Search	4
`127933-IVORY`	Ivory (41)	13,999	11,190	Not Visible Individually	4
`127933-RED`	Red (43)	15,999	12,799	Not Visible Individually	4

6.2 Database tables involved

`catalog_product_super_attribute`	Which attribute drives the configurable (here color, id 93)
`catalog_product_super_link`	Parent ↔ child links
`catalog_product_relation`	Magento's internal parent/child index
`catalog_product_entity_int`	Each child's color option_id
`catalog_product_entity_decimal`	Each child's price and special_price
`eav_attribute_option_swatch`	**The swatch colours** — option_id, store_id, type, hex value

Swatch rows currently stored

swatch_id	option_id	Label	type	Hex
1	40	Beige	1	`#d7c9ae`
2	42	Cream	1	`#f2e8d5`
3	4	Grey	1	`#8c8c8c`
4	41	Ivory	1	`#efe7d8`
5	43	Red	1	`#d32f2f`

`type = 1` means a visual colour swatch (2 would be an image, 0 a text swatch).

6.3 How the swatch conversion was done

Magento's `Swatches\Model\Plugin\EavAttribute` expects `optionvisual` / `swatchvisual` arrays shaped like an admin form post, and feeding it those would have rewritten the existing options. Instead the patch writes the two things that actually define a swatch:

• the `swatch_input_type` flag inside the attribute's `additional_data` JSON — what `SwatchAttributeType::isVisualSwatch()` reads;
• one row per option in `eav_attribute_option_swatch`.

This is idempotent and leaves option IDs already assigned to products untouched.

6.4 Two bugs found and fixed

Swatch clicks did not change the gallery

Magento's swatch renderer switches images through Fotorama: `updateBaseImage()` looks for a gallery widget on `[data-gallery-role=gallery-placeholder]` and, finding none, subscribes to a `gallery:loaded` event and returns. This theme renders its own static grid instead of Fotorama, so that event never fired and images never changed. Prices still updated because that is a separate code path.

Fixed with `web/js/swatch-gallery-mixin.js`, which overrides `updateBaseImage` to rebuild the theme's grid from the same `jsonConfig.images` data, and falls back to `_super()` anywhere the custom gallery is absent.

The discount badge disappeared

Configurables render through `Magento_ConfigurableProduct`'s own price template, which the theme had never overridden — so the custom *(-20% OFF)* badge was replaced by Magento's plain “As low as”. A theme override restores it, and `web/js/discount-badge.js` recomputes the percentage on the `priceUpdated` event, because the server only renders the badge for the cheapest variant.

6.5 Files

Layer	Path
Patch	`Setup/Patch/Data/ConfigureColorSwatches.php`
Patch	`Setup/Patch/Data/ConvertNiecelToConfigurable.php`
Patch	`Setup/Patch/Data/FixSofaVariantImages.php`
JS mixin	`THEME/web/js/swatch-gallery-mixin.js`
JS	`THEME/web/js/discount-badge.js`
Template	`THEME/Magento_ConfigurableProduct/templates/product/price/final_price.phtml`
Config	`THEME/requirejs-config.js`

6.6 How to test

Open `/ae_en/niecel-sectional-sofa-127933` in a browser and click each colour chip. Expected behaviour:

Swatch	Gallery	Price	Was	Badge
Ivory	4 ivory sofa photos	11,190	13,999	(-20% OFF)
Red	4 red sofa photos	12,799	15,999	(-20% OFF)

Switching back and forth must restore each colour's own images. Verify the links in the database:

```
SELECT p.sku AS parent, c.sku AS child
FROM catalog_product_super_link l
JOIN catalog_product_entity p ON p.entity_id = l.parent_id
JOIN catalog_product_entity c ON c.entity_id = l.product_id;
```

7. Second website — India

7.1 What was implemented

A second storefront for India sharing the UAE catalogue and theme, with its own currency, locale and independently managed prices.

Level	UAE	India
Website	id 1, code `base`	id 2, code `in`
Store group	`main_website_store`, root cat 2	`in_store`, root cat 2
Store view	id 1, `ae_en`	id 2, `in_en`
Currency	AED	INR
Locale	en_US	en_IN
URL	`midhun2.test/ae_en/`	`midhun2.test/in_en/`

Tables involved

`store_website`	The website row
`store_group`	The store group, including its root_category_id
`store`	The store view
`core_config_data`	Scoped configuration (scope = websites / stores, scope_id = that row's id)
`url_rewrite`	Per-store URL rewrites — **store_id is part of every row**

7.2 Per-website pricing

`catalog/price/scope` was changed to **Website**.
`Catalog\Model\Product\Attribute\Backend\Price::setScope()` reads that flag and re-scopes the price attribute at runtime, so no EAV rows had to be rewritten. Prices then live per store in `catalog_product_entity_decimal`:

SKU	store_id 0 — AED (UAE inherits)	store_id 2 — INR (India)
`130917`	4,495 → 3,575	102,999 → 81,999
`127933-IVORY`	13,999 → 11,190	319,999 → 255,999
`127933-RED`	15,999 → 12,799	364,999 → 291,999

The INR figures are a placeholder conversion at a flat rate, rounded to a trailing 999 and keeping the same 20% discount. **Replace them with real commercial pricing.**

7.3 Three things that needed handling

Store code in URLs is global	`web/url/use_store` is `showInDefault` only, and `Store::isUseStoreInUrl()` reads the default scope, so it prefixes *every* view. The original view was therefore renamed from `default` to `ae_en` — otherwise the UAE site would answer on /default/. Bare URLs without a code still resolve to the default store.
URL rewrites are per store	Magento builds rewrites when a category or product is saved, so a store view added after the catalogue exists starts with none and every category URL 404s. `GenerateRegionalUrlRewrites` regenerates them for all views.

en_IN is not an allowed locale	`Framework\Locale\Config` ships only en_AU, en_CA, en_GB, en_IE, en_NZ and en_US. Magento generated /static/.../en_IN/ asset URLs that its own static handler then rejected, so the India storefront rendered with no CSS. `etc/di.xml` adds en_IN to the list — ICU supports it, and it groups in lakhs the way India writes prices.

7.4 Theme changes for multi-region

Change	Why
`PriceFormatter` made currency-aware	The dirham glyph was hardcoded; it now applies only when the store currency is AED
Home card prices use it	They previously printed the AED glyph next to rupee amounts
BNPL block hidden outside AED	Tabby and Tamara are GCC-only providers — wrong in India commercially, not just visually
Header globe switcher made dynamic	Built from the live store list via `RegionSwitcher`, so both regions are reachable

7.5 Files

Layer	Path
Patch	`Setup/Patch/Data/AddIndiaWebsite.php`
Patch	`Setup/Patch/Data/ConfigureIndiaStore.php`
Patch	`Setup/Patch/Data/SetIndiaCatalogPrices.php`
Patch	`Setup/Patch/Data/RestoreRegionalPrices.php`
Patch	`Setup/Patch/Data/ConfigureUaeRegion.php`
Patch	`Setup/Patch/Data/GenerateRegionalUrlRewrites.php`
ViewModel	`ViewModel/RegionSwitcher.php`
ViewModel	`ViewModel/PriceFormatter.php  (modified)`
DI	`etc/di.xml  (en_IN locale)`
Theme	`Magento_Theme/templates/html/header/account.phtml`
Theme	`Magento_Cms/templates/home/products.phtml`
Theme	`Magento_Catalog/templates/product/view/bnpl.phtml`

7.6 How to test

Check	UAE	India
Home	`/ae_en/`	`/in_en/`
PDP	`/ae_en/niecel-sectional-sofa-127933`	`/in_en/niecel-sectional-sofa-127933`
Category	`/ae_en/furniture/sofa-seating/corner-sofa`	`/in_en/...`
Currency shown	AED glyph	■ with lakh grouping
BNPL cards	visible	hidden

The globe icon in the header should list both regions and link to the other one. Confirm the price scoping in the database:

```
SELECT e.sku, a.attribute_code, d.store_id, d.value
FROM catalog_product_entity_decimal d
JOIN eav_attribute a ON a.attribute_id = d.attribute_id
JOIN catalog_product_entity e ON e.entity_id = d.entity_id
WHERE a.attribute_code IN ('price','special_price')
ORDER BY e.sku, d.store_id;
```

8. Custom table — product enquiries

8.1 What was implemented

A “Request a quote” form on every product page, storing leads in a custom table, with a full admin grid and edit form behind an ACL resource.

8.2 The table

Created with **declarative schema** — `etc/db_schema.xml` plus a generated `db_schema_whitelist.json`, applied by `setup:upgrade`. The old `InstallSchema` / `UpgradeSchema` classes are deprecated and were not used.

```
CREATE TABLE `codilar_twoxl_product_enquiry` (
  `entity_id`  int unsigned      NOT NULL AUTO_INCREMENT,
  `name`       varchar(120)      NOT NULL,
  `email`      varchar(180)      NOT NULL,
  `phone`      varchar(40)       DEFAULT NULL,
  `sku`        varchar(64)       DEFAULT NULL,
  `message`    text,
  `status`     smallint unsigned NOT NULL DEFAULT '1',
  `store_id`   smallint unsigned DEFAULT NULL,
  `created_at` timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP,
  `updated_at` timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
                          ON UPDATE CURRENT_TIMESTAMP,
  PRIMARY KEY (`entity_id`),
  KEY CODILAR_TWOXL_PRODUCT_ENQUIRY_STATUS     (`status`),
  KEY CODILAR_TWOXL_PRODUCT_ENQUIRY_SKU        (`sku`),
  KEY CODILAR_TWOXL_PRODUCT_ENQUIRY_EMAIL      (`email`),
  KEY CODILAR_TWOXL_PRODUCT_ENQUIRY_CREATED_AT (`created_at`),
  CONSTRAINT CODILAR_TWOXL_PRODUCT_ENQUIRY_STORE_ID_STORE_STORE_ID
    FOREIGN KEY (`store_id`) REFERENCES `store` (`store_id`)
    ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

Design decisions

4 indexes	status, sku, email and created_at — the columns the admin grid filters and sorts on
`ON DELETE SET NULL`	Chosen over the more common CASCADE: deleting a store view should not destroy sales leads. This requires `store_id` to be nullable.
status as smallint	1 new, 2 in progress, 3 closed — constants on the model, labels from a source model
store_id recorded	Enquiries are region-aware, so UAE and India leads can be told apart

8.3 Code around the table

Layer	File	Role
Schema	`etc/db_schema.xml`	Table definition
Schema	`etc/db_schema_whitelist.json`	Generated; permits later alters
Model	`Model/ProductEnquiry.php`	Status constants, getStatusLabel()
Resource	`Model/ResourceModel/ProductEnquiry.php`	Table binding
Collection	`Model/ResourceModel/ProductEnquiry/Collection.php`	Plus addOpenFilter()
Source	`Model/Source/EnquiryStatus.php`	Dropdown options
Storefront	`Controller/Enquiry/Save.php`	Validates and stores a submission
Admin	`Controller/Adminhtml/Enquiry/*.php`	Index, Edit, Save, Delete

Layer	File	Role
Admin UI	`view/adminhtml/ui_component/..._listing.xml`	Grid with filters
Admin UI	`view/adminhtml/ui_component/..._form.xml`	Edit form
Admin UI	`Ui/Component/Listing/Column/EnquiryActions.php`	Row Edit / Delete links
Admin UI	`Model/ProductEnquiry/DataProvider.php`	Feeds the edit form
Access	`etc/acl.xml, etc/adminhtml/{routes,menu}.xml`	Permission, route, menu entry
Theme	`Magento_Catalog/templates/product/view/enquiry.phtml`	The storefront form
Seed	`Setup/Patch/Data/SeedProductEnquiries.php`	Two sample rows

8.4 How to test

Storefront — must be a real browser

Open `/ae_en/niecel-sectional-sofa-127933`, expand **Request a quote**, fill Name and Email, submit. A green confirmation appears and a row is created. Try an empty name or an invalid email to see the validation errors.

This cannot be tested with curl. The page is full-page cached, so the `form_key` in the HTML is stale; `Magento_PageCache/js/form-key-provider` replaces it client side. Without that JS, Magento's CSRF check rejects the POST before the controller runs — you get a 302 back to the product page and no row, which looks like a silent failure. Look for *Request validation failed for action* in `var/log/debug.log`.

Admin

Catalog → **Product Enquiries**. Filter by status, store view or date; use Edit to change a status; use Delete (it asks for confirmation).

Database

```
SELECT entity_id, name, email, sku, status, store_id, created_at
FROM codilar_twoxl_product_enquiry ORDER BY entity_id DESC;
```

8.5 Changing the table later

Never alter the table by hand. Edit `db_schema.xml`, regenerate the whitelist, then upgrade — Magento diffs the declaration against the live table and applies only the difference. Anything absent from the whitelist is left alone.

```
php bin/magento setup:db-declaration:generate-whitelist --module-name=Codilar_Twoxl
php bin/magento setup:upgrade
```

9. Operational notes and gotchas

These cost real debugging time on this install and are worth knowing before changing anything.

9.1 Caches

`cache:flush` alone is sometimes **not enough**. Stale data has survived it twice here: once as an outdated configurable `jsonConfig` (old variant images kept rendering after the database was already correct) and once as a corrupted UI-component entry that made the admin enquiry grid throw *Undefined array key "arguments"*. The reliable sequence:

```
php bin/magento cache:clean
php bin/magento cache:flush
rm -rf var/cache/* var/page_cache/*
```

9.2 Static content

There is no compiled CSS on disk — this install generates static files on demand through `static.php`. After changing LESS or JS:

```
rm -rf pub/static/frontend/Codilar/twoxl var/view_preprocessed/*
```

If the site is ever switched to production mode, on-demand generation stops and both locales must be deployed explicitly: `setup:static-content:deploy en_US en_IN`.

9.3 Writing data patches that touch existing products

A patch that **updates** an existing product will fail with *Area code is not set*. `Product::getIdentities()` reads the area code unconditionally, and `PatchApplier` defers every save's commit callbacks to its own outer `commit()` — which happens after `apply()` has returned and `emulateAreaCode()` has restored the unset area. Patches that only *create* products do not hit this.

The fix is the framework's own opt-out: implement `NonTransactionableInterface`, so each save commits inside the emulation. Used by `FixSofaVariantImages`, `SetIndiaCatalogPrices` and `RestoreRegionalPrices`.

9.4 Config written in a patch is not visible to the same patch run

`SetIndiaCatalogPrices` originally wrote the rupee prices while the cached config still reported a *global* price scope. The store-scoped write collapsed onto the default row and overwrote the dirham prices. Any patch that writes configuration and then acts on it must call `ReinitableConfigInterface::reinit()` first.

9.5 Error reports

Storefront and admin errors show a record number rather than a trace. The matching file is `var/report/<id>` — JSON containing the URL and the full stack trace. Start there, not in the browser.

9.6 Known gaps

Item	Status
INR prices	Placeholder conversion — replace with commercial pricing
Sofa dimensions and copy	Invented placeholders; verify against the real product data
Ivory / parent images	Two copies of the same photos exist (admin upload plus fixture import)
`services.phtml`	Still reads “orders above AED 500” on both sites
India payment methods	None configured

Item	Status
AED ↔ INR rate	Not set — not needed for storefront pricing, but admin cross-website reporting will want one
Enquiry notifications	No email is sent when an enquiry arrives
Enquiry mass actions	No bulk delete or bulk status change on the grid

Appendix A — File inventory

A.1 Module — data patches

File	Purpose
`AddProductAttributes.php`	PDP specification attributes
`ConfigureStore.php`	AED currency, suffix-less URLs, titles
`CreateTresaDiningTable.php`	Menu categories and the dining table
`ConfigureColorAttribute.php`	Shows color on the PDP
`AddTableMaterialAttribute.php`	Relocated from Vendor module, with getAliases()
`AddSofaAttributes.php`	Four sofa dropdowns, tops up seating_capacity and color
`CreateNiecelSectionalSofa.php`	Sofa categories and the product
`AddCategoryBannerAttribute.php`	category_banner on catalog_category
`ConfigureColorSwatches.php`	Adds Red; converts color to visual swatches
`ConvertNiecelToConfigurable.php`	Colour children and the configurable parent
`FixSofaVariantImages.php`	Real ivory photos on the parent and Ivory child
`AddIndiaWebsite.php`	India website, group, view; renames default to ae_en
`ConfigureIndiaStore.php`	Price scope, store code in URLs, INR, en_IN
`SetIndiaCatalogPrices.php`	Website assignment and INR prices
`RestoreRegionalPrices.php`	Re-pins AED and INR to the right scopes
`ConfigureUaeRegion.php`	Country AE for the UAE website
`GenerateRegionalUrlRewrites.php`	Rebuilds rewrites for every store view
`SeedProductEnquiries.php`	Two sample enquiries

A.2 Module — other code

File	Purpose
`etc/db_schema.xml`	codilar_twoxl_product_enquiry table
`etc/db_schema_whitelist.json`	Generated schema whitelist
`etc/di.xml`	en_IN locale, enquiry grid data source
`etc/acl.xml`	Codilar_Twoxl::enquiry permission
`etc/adminhtml/routes.xml`	Admin front name twoxl_admin
`etc/adminhtml/menu.xml`	Catalog -> Product Enquiries
`Model/ProductEnquiry.php`	Enquiry model
`Model/ResourceModel/ProductEnquiry.php`	Resource
`Model/ResourceModel/ProductEnquiry/Collection.php`	Collection
`Model/ProductEnquiry/DataProvider.php`	Admin form data
`Model/Source/EnquiryStatus.php`	Status options
`Controller/Enquiry/Save.php`	Storefront submission

File	Purpose
`Controller/Adminhtml/Enquiry/{Index,Edit,Save,Delete}.php`	Admin actions
`Ui/Component/Listing/Column/EnquiryActions.php`	Grid row links
`ViewModel/CategoryBanner.php`	Resolves the banner URL
`ViewModel/RegionSwitcher.php`	Header country list
`ViewModel/PriceFormatter.php`	Currency-aware formatting (modified)
`view/adminhtml/ui_component/category_form.xml`	Adds the banner uploader
`view/adminhtml/ui_component/codilar_twoxl_enquiry_listing.xml`	Admin grid
`view/adminhtml/ui_component/codilar_twoxl_enquiry_form.xml`	Admin edit form
`view/adminhtml/layout/twoxl_admin_enquiry_{index,edit}.xml`	Admin layouts

A.3 Theme

File	Purpose
`requirejs-config.js`	Registers price-box and swatch-gallery mixins
`web/js/swatch-gallery-mixin.js`	Makes swatches switch the custom gallery
`web/js/discount-badge.js`	Recomputes the discount badge on price change
`Magento_ConfigurableProduct/.../final_price.phtml`	Restores the badge on configurables
`Magento_Catalog/layout/catalog_category_view.xml`	Category banner block
`Magento_Catalog/templates/category/banner.phtml`	Category banner markup
`Magento_Catalog/templates/product/view/enquiry.phtml`	Request a quote form
`Magento_Catalog/templates/product/view/bnpl.phtml`	BNPL hidden outside AED (modified)
`Magento_Cms/templates/home/products.phtml`	Currency-aware card prices (modified)
`Magento_Theme/templates/html/header/account.phtml`	Dynamic region switcher (modified)
`Magento_Theme/layout/default.xml`	Injects RegionSwitcher (modified)
`web/css/source/twoxl/_category.less`	Category banner styles
`web/css/source/twoxl/_enquiry.less`	Enquiry form styles
`web/css/source/_extend.less`	Imports every partial (modified)

Appendix B — Applied patches, in order

As recorded in the `patch_list` table. Patch 194 is the original Vendor-module entry kept so the relocated class in 197 is not re-applied.

#	Patch
189	`AddProductAttributes`
190	`ConfigureStore`
192	`CreateTresaDiningTable`
193	`ConfigureColorAttribute`
194	`Vendor\FurnitureAttributes\...\AddTableMaterialAttribute (legacy name)`
195	`AddSofaAttributes`
196	`CreateNiecelSectionalSofa`
197	`AddTableMaterialAttribute (relocated)`
198	`AddCategoryBannerAttribute`
199	`ConfigureColorSwatches`
200	`ConvertNiecelToConfigurable`
201	`FixSofaVariantImages`
202	`AddIndiaWebsite`
203	`ConfigureIndiaStore`
204	`SetIndiaCatalogPrices`
205	`RestoreRegionalPrices`
206	`ConfigureUaeRegion`
207	`GenerateRegionalUrlRewrites`
208	`SeedProductEnquiries`

Appendix C — Quick command reference

Apply all patches	`php bin/magento setup:upgrade`
Full cache clear	`cache:clean && cache:flush && rm -rf var/cache/* var/page_cache/*`
Rebuild static	`rm -rf pub/static/frontend/Codilar/twoxl var/view_preprocessed/*`
Reindex	`php bin/magento indexer:reindex`
Admin URL	`php bin/magento info:adminuri`
List stores	`php bin/magento store:list`
Regenerate schema whitelist	`setup:db-declaration:generate-whitelist --module-name=Codilar_Twoxl`